

Sharp Poisson asymptotics for the longest arithmetic progression in a random set

An entropy coefficient, a second-order correction,
and the lattice oscillation of the optimal error

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Abstract

Let U_n be the longest arithmetic progression in the Bernoulli- p random subset of $[n]$, with $p \in (0, 1)$ fixed. We study the accuracy of the Poisson approximation to $\mathbb{P}(U_n < r)$, rather than only its limiting law. Writing $M_r(n)$ for the number of r -term progressions, the exact mean of progression heads is $\lambda_{n,r} = M_r(n)p^r - M_{r+1}(n)p^{r+1}$. We give a self-contained argument for the uniform expansion

$$\mathbb{P}(U_n < r) = e^{-\lambda_{n,r}} \left(1 + c_p \lambda_{n,r}^2 \frac{r^2}{n} \right) + O_p \left(\frac{(\log n)^{3/2}}{n} \right), \quad 2 \leq r \leq n,$$

where $c_p = \frac{1-p}{p}(\frac{5}{3} - \frac{\pi^2}{9})$. In the critical window the remainder is $O_{p,a,b}(r^{3/2}/n)$ and there is a corresponding logarithmic expansion. Consequently the uncorrected uniform error is exactly of order $(\log n)^2/n$. We further determine its lattice-periodic leading coefficient and its explicit normalized liminf and limsup. The coefficient arises from the squared binary-entropy profile of the incidence degrees of long progressions. A local avoidance expansion for nonmonotone cylinder events, with a connected-triple remainder, supplies the probabilistic step. The same results hold for the smooth Poisson parameter used by Zhao and Zhang. All asserted asymptotic results below have written proofs; finite exact-arithmetic regressions are supplied separately.

Status and scope. This is an unrefereed research manuscript, not a Lean-verified result. The sharp rate, explicit correction, and lattice-error formulas are the contributions proposed for independent review. The literature search does not certify priority. The argument does not assume the correctness of any new manuscript in either linked repository, and it does not improve a deterministic Szemerédi or van der Waerden bound.

Contents

1	Problem, results, and relation to existing work	3
1.1	The random sequence	3
1.2	Main statements	3
1.3	What is inherited, and what is being added	4
2	Progression heads and the exact mean	5
3	A local expansion for avoidance of cylinder events	5
3.1	Parameters and elementary conditional bounds	6
3.2	An exponentially damped first-order estimate	7
3.3	The second factorial cumulant, with a triple remainder	7
4	Geometry of pairs and connected triples	9
4.1	Pair counts and compatible overlaps	9
4.2	A triple estimate adequate for a second-order law	9
5	The entropy profile of progression incidence	10
6	Computing the leading covariance term	11
7	Assembly of the critical and uniform expansions	12
7.1	A logarithmic band with a weighted remainder	12
7.2	Why the tails do not lose logarithms	13
8	The optimal error has an explicit periodic amplitude	14
9	A separate sparse-density consequence	15
10	Verification, limitations, and integration	16
10.1	What the executable artifact actually checks	16
10.2	Claims that are not being made	16
10.3	Repository placement and proof dependencies	17
11	Further research questions	17

1 Problem, results, and relation to existing work

1.1 The random sequence

Fix $p \in (0, 1)$ and put $q = 1 - p$ and $\alpha = -\log p > 0$. Let $\xi_1, \dots, \xi_n$ be independent Bernoulli- p variables. The random set is $A_n = \{x \in [n] : \xi_x = 1\}$. An arithmetic progression always has positive integer common difference. Let U_n be its largest possible length in A_n , with $U_n = 0$ for the empty set and $U_n = 1$ when there is no two-term progression but the set is nonempty. For $2 \leq r \leq n$, write

$$D_r = \left\lfloor \frac{n-1}{r-1} \right\rfloor, \quad M_r(n) = \sum_{d=1}^{D_r} (n - (r-1)d) = nD_r - \frac{(r-1)D_r(D_r+1)}{2}, \quad (1)$$

$$\lambda_{n,r} = p^r M_r(n) - p^{r+1} M_{r+1}(n), \quad (2)$$

$$\bar{\lambda}_{n,r} = \frac{n^2 p^r (p + qr)}{2r(r-1)}. \quad (3)$$

The definition of M_r also applies when $r > n$, giving zero. All logarithms are natural. The notation O_p here denotes a deterministic implicit constant depending on p , not stochastic boundedness. Such constants may blow up as $p \downarrow 0$ or $p \uparrow 1$.

The key positive constants are

$$H = \int_0^1 (-u \log u - (1-u) \log(1-u))^2 du = \frac{5}{6} - \frac{\pi^2}{18}, \quad c_p = \frac{2qH}{p} = \frac{q}{p} \left(\frac{5}{3} - \frac{\pi^2}{9} \right). \quad (4)$$

In particular, $H \approx 0.285021977717258$ and $c_{1/2} \approx 0.570043955434516$.

1.2 Main statements

Theorem 1.1 (Uniform second-order approximation). *For every fixed $p \in (0, 1)$ and either choice $\eta_{n,r} = \lambda_{n,r}$ or $\eta_{n,r} = \bar{\lambda}_{n,r}$,*

$$\sup_{2 \leq r \leq n} \left| \mathbb{P}(U_n < r) - e^{-\eta_{n,r}} \left(1 + c_p \eta_{n,r}^2 \frac{r^2}{n} \right) \right| = O_p \left(\frac{(\log n)^{3/2}}{n} \right). \quad (5)$$

The polynomial correction in this statement is intentional. Extending the exponential $\exp(-\eta + c_p \eta^2 r^2/n)$ to very small r would be incorrect: far outside the critical window its positive quadratic exponent can become enormous.

Theorem 1.2 (Critical-window logarithmic expansion). *Fix $0 < a < b < \infty$. Uniformly over integers r satisfying $a \leq \lambda_{n,r} \leq b$,*

$$r = \frac{2 \log n - \log \log n + O_{p,a,b}(1)}{\alpha}, \quad (6)$$

$$\log \mathbb{P}(U_n < r) = -\lambda_{n,r} + c_p \lambda_{n,r}^2 \frac{r^2}{n} + O_{p,a,b} \left(\frac{r^{3/2}}{n} \right), \quad (7)$$

$$\mathbb{P}(U_n < r) - e^{-\lambda_{n,r}} = c_p e^{-\lambda_{n,r}} \lambda_{n,r}^2 \frac{r^2}{n} + O_{p,a,b} \left(\frac{r^{3/2}}{n} \right). \quad (8)$$

The same conclusions hold with $\bar{\lambda}$ in place of λ , including when the critical-window condition is imposed on $\bar{\lambda}$.

Thus the Poisson estimate has a strictly positive leading bias in the critical window: the true avoidance probability is larger.

Theorem 1.3 (Optimal error and its lattice oscillation). *For either parameter η in Theorem 1.1, let*

$$\mathcal{E}_n(\eta) = \sup_{2 \leq r \leq n} |\mathbb{P}(U_n < r) - e^{-\eta_{n,r}}|, \quad t_n = \frac{2 \log n - \log \log n + \log(q\alpha/4)}{\alpha},$$

and define the continuous one-periodic function

$$\Phi_p(\theta) = \max_{j \in \mathbb{Z}} \left\{ e^{2\alpha(\theta-j)} \exp(-e^{\alpha(\theta-j)}) \right\}. \quad (9)$$

Then

$$\mathcal{E}_n(\eta) = \frac{4c_p}{\alpha^2} \frac{(\log n)^2}{n} \Phi_p(\{t_n\}) + O_p\left(\frac{(\log n)^{3/2}}{n}\right). \quad (10)$$

In particular $\mathcal{E}_n(\eta) = \Theta_p((\log n)^2/n)$, and

$$\limsup_{n \rightarrow \infty} \frac{n\mathcal{E}_n(\eta)}{(\log n)^2} = \frac{16c_p}{\alpha^2 e^2}, \quad (11)$$

$$\liminf_{n \rightarrow \infty} \frac{n\mathcal{E}_n(\eta)}{(\log n)^2} = \frac{16c_p}{q^2} \exp\left(-\frac{2\alpha}{q}\right). \quad (12)$$

These two constants are distinct for every fixed $p \in (0, 1)$.

1.3 What is inherited, and what is being added

Benjamini, Yadin, and Zeitouni studied longest progressions in random subsets using Poisson approximation [1]. Zhao and Zhang's Theorem 1.1(1), in the preprint inspected here, gives

$$\sup_{2 \leq r \leq n} |\mathbb{P}(U_n < r) - e^{-\bar{\lambda}_{n,r}}| = O_p\left(\frac{\log^4 n \log \log n}{n}\right). \quad (13)$$

Their progression-head construction and overlap estimates are directly relevant [6]. We explicitly reuse and reprove those ingredients; neither the construction nor the three-term pair estimate below is claimed as new. Theorems 1.1–1.3 refine the displayed estimate (13), rather than merely substitute constants in its proof.

Higher-order avoidance estimates through joint cumulants are an established subject. In particular, Mousset, Noever, Panagiotou, and Samotij give systematic expansions for binomial-subset nonexistence probabilities, including fixed-length arithmetic progressions [2]. Our event family is nonmonotone because a progression head requires a zero immediately before a string of ones. Section 3 therefore proves the particular local expansion needed here directly. We make no priority claim for the general principle that the second factorial cumulant supplies the first Poisson correction.

The ProveIt Ramsey directory contains research branches devoted to Gowers' proof, quasipolynomial progressions, square differences, and van der Waerden problems [5]. The OpenAI collection includes an arithmetic- progression manuscript family and explicitly distinguishes verification statuses [3, 4]. These motivate the choice of topic, but are not premises of any theorem proved here. This paper belongs in probabilistic Ramsey/additive combinatorics, not in the dependency chain of a deterministic density-increment proof.

2 Progression heads and the exact mean

Let

$$\mathcal{A}_{n,r} = \{(a, d) : a, d \in \mathbb{Z}_{>0}, a + (r-1)d \leq n\}.$$

For $i = (a, d)$, define the positive core and support

$$C_i = \{a, a+d, \dots, a+(r-1)d\}, \quad S_i = C_i \cup (\{a-d\} \cap [n]).$$

The head event is

$$E_i = \begin{cases} \{\xi_x = 1 \text{ for all } x \in C_i\}, & a \leq d, \\ \{\xi_{a-d} = 0\} \cap \{\xi_x = 1 \text{ for all } x \in C_i\}, & a > d, \end{cases} \quad Y = \sum_{i \in \mathcal{A}_{n,r}} \mathbf{1}_{E_i}. \quad (14)$$

A boundary head has $a \leq d$ and probability p^r ; an interior head has probability qp^r .

Lemma 2.1 (Exact reduction). *For every $n \geq 2$ and $2 \leq r \leq n$,*

$$\{U_n < r\} = \{Y = 0\}, \quad \mathbb{E}Y = \lambda_{n,r}.$$

Moreover $Y_{n,r+1} \leq Y_{n,r}$ pointwise, and therefore $\lambda_{n,r}$ is nonincreasing in r . Uniformly in $2 \leq r \leq n$,

$$|\lambda_{n,r} - \bar{\lambda}_{n,r}| \leq 2np^r. \quad (15)$$

If $r = O(\log n)$ and $r \rightarrow \infty$, then

$$\lambda_{n,r} = \frac{qn^2p^r}{2r} \left(1 + O_p\left(\frac{1}{r} + \frac{r}{n}\right)\right). \quad (16)$$

Proof. Any selected r -term progression can be extended backwards along its common difference until the first selected element of that run is reached. The resulting run begins at a head and still has length at least r . Conversely every head supplies an r -term progression. The same starting point and difference of an $(r+1)$ -head also define an r -head, proving pointwise monotonicity.

There are $M_r(n)$ possible cores. The interior ones correspond bijectively to $(r+1)$ -term progressions by adjoining the predecessor. Replacing their probability p^r by qp^r proves (2). For $m = r-1$, put $u = n/m$ and $t = D_r$ in (1). We have $0 < u - t \leq 1$, and a direct expansion gives

$$M_r(n) = \frac{n^2}{2m} - \frac{n}{2} + \frac{m}{2}((u-t) - (u-t)^2).$$

Thus $|M_r(n) - n^2/(2(r-1))| \leq n$; the same bound holds for M_{r+1} . This proves (15). Combining it with the exact formula (3) proves (16). $\square$

3 A local expansion for avoidance of cylinder events

This section is independent of arithmetic progressions. Its purpose is to justify the passage from a covariance calculation to a zero-probability asymptotic, with a quantified remainder. It does not assume positivity of correlations or monotonicity of the events.

3.1 Parameters and elementary conditional bounds

Let $(E_i)_{i \in I}$ be a finite family of events, each determined by a specified finite set of independent coordinates S_i . Join $i \neq j$ when $S_i \cap S_j \neq \emptyset$, write $i \sim j$, and let $B_i = \{i\} \cup \{j : j \sim i\}$. Set

$$\begin{aligned} \pi_i &= \mathbb{P}(E_i), \quad \pi_{ij} = \mathbb{P}(E_i \cap E_j), \quad \pi_{ijk} = \mathbb{P}(E_i \cap E_j \cap E_k), \\ \lambda &= \sum_i \pi_i, \quad \delta_i = \sum_{j \in B_i} \pi_j, \quad \delta = \max_i \delta_i, \\ b_1 &= \sum_i \pi_i \delta_i, \quad b_2 = \sum_i \sum_{j \sim i} \pi_{ij}, \end{aligned} \tag{17}$$

$$T = \sum_{\substack{\{i,j,k\} \subset I \\ \text{induced graph connected}}} \pi_{ijk}. \tag{18}$$

The pairs in b_2 are ordered; the triples in T are unordered and distinct. Write $F_A = \bigcap_{j \in A} E_j^c$.

Lemma 3.1 (Conditional control). *Suppose $\delta \leq 1/8$. Every F_A has positive probability, and for $i \notin A$,*

$$\mathbb{P}(E_i \mid F_A) \leq 2\pi_i. \tag{19}$$

More generally, let G be an event determined by the union of at most m of the supports S_i . Let N_G be the union of the closed neighborhoods of those at most m supports and put $w = \sum_{j \in A \cap N_G} \pi_j$. Then

$$\mathbb{P}(G \mid F_A) \leq \mathbb{P}(G) e^{4w}, \tag{20}$$

$$(\mathbb{P}(G \mid F_A) - \mathbb{P}(G))_+ \leq 4e^{4m\delta} \mathbb{P}(G) w, \tag{21}$$

$$(\mathbb{P}(G) - \mathbb{P}(G \mid F_A))_+ \leq e^{4(m+1)\delta} \sum_{j \in A \cap N_G} \mathbb{P}(G \cap E_j). \tag{22}$$

Proof. We give the usual local-lemma induction to fix the conditional statement. For $A = \emptyset$, (19) holds. Assume the conditional bound for conditioning sets of size less than $|A|$. First obtain positivity of F_A by removing one index, using that bound, and observing that $1 - 2\pi_i > 0$. Then split A into the neighbors $V = A \cap B_i$ and the nonneighbors $W = A \setminus B_i$. When $V = \emptyset$, independence gives the claim. Otherwise

$$\mathbb{P}(E_i \mid F_A) \leq \frac{\mathbb{P}(E_i \mid F_W)}{\mathbb{P}(F_V \mid F_W)} \leq \frac{\pi_i}{\prod_{j \in V} (1 - 2\pi_j)} \leq \frac{\pi_i}{1 - 2 \sum_{j \in V} \pi_j} \leq 2\pi_i.$$

The product bound uses the induction hypothesis, adding the elements of V one at a time; every intermediate conditioning set has smaller size. Positivity at the next size follows by adding one event complement and using $2\pi_i \leq 1/4 < 1$.

For G , split A into $V = A \cap N_G$ and $W = A \setminus N_G$. The event G is independent of all coordinates used by F_W . The same product argument and $-\log(1 - 2x) \leq 4x$ for $0 \leq x \leq 1/8$ give

$$\mathbb{P}(G \mid F_A) \leq \frac{\mathbb{P}(G)}{\prod_{j \in V} (1 - 2\pi_j)} \leq \mathbb{P}(G) e^{4w}.$$

Since $w \leq m\delta$, $e^{4w} - 1 \leq 4we^{4m\delta}$ proves the upper excess estimate. For the lower estimate, the denominator $\mathbb{P}(F_V \mid F_W)$ is at most one, so the union bound yields

$$\mathbb{P}(G \mid F_A) \geq \mathbb{P}(G) - \sum_{j \in V} \mathbb{P}(G \cap E_j \mid F_W).$$

Each intersection uses at most $m + 1$ original supports. Applying the just-proved upper estimate proves (22). $\square$

3.2 An exponentially damped first-order estimate

Lemma 3.2 (Coarse logarithmic sandwich). *Under $\delta \leq 1/8$,*

$$-\lambda - 4b_1 \leq \log \mathbb{P}(F_I) \leq -\lambda + 2b_2. \quad (23)$$

If also $b_2 \leq \lambda/4$, then

$$|\mathbb{P}(F_I) - e^{-\lambda}| \leq (4b_1 + 2b_2)e^{-\lambda/2}. \quad (24)$$

Proof. Order I , and put $z_i = \mathbb{P}(E_i \mid F_{\{j:j < i\}})$ and $u_i = \sum_{j < i, j \sim i} \pi_j$. Lemma 3.1 gives

$$z_i \leq \pi_i e^{4u_i}, \quad z_i \geq \pi_i - e^{8\delta} \sum_{j < i, j \sim i} \pi_{ij}.$$

The second inequality summed over i , followed by $\log(1 - z) \leq -z$, gives the upper bound because $e^{8\delta}/2 \leq e/2 < 2$. For the lower bound,

$$\sum_i (z_i - \pi_i) \leq 4e^{4\delta} \sum_i \pi_i u_i \leq 4 \left(b_1 - \sum_i \pi_i^2 \right).$$

Here $\sum_i \pi_i u_i = (b_1 - \sum_i \pi_i^2)/2$ and $2e^{1/2} < 4$. As $z_i \leq 2\pi_i \leq 1/4$, we may use $\log(1 - z_i) \geq -z_i - z_i^2$ and $\sum_i z_i^2 \leq 4 \sum_i \pi_i^2$. This proves (23). Finally write $\mathbb{P}(F_I) = e^{-\lambda+v}$ with $-4b_1 \leq v \leq 2b_2$. The elementary inequality $|e^v - 1| \leq |v|e^{\max(v,0)}$ proves (24). $\square$

3.3 The second factorial cumulant, with a triple remainder

Proposition 3.3 (Second-order local avoidance). *There is an absolute constant C such that, whenever $\delta \leq 1/8$,*

$$\left| \log \mathbb{P}(F_I) + \lambda - \frac{\text{Var}(\sum_i \mathbf{1}_{E_i}) - \lambda}{2} \right| \leq C(\delta(b_1 + b_2) + T). \quad (25)$$

Proof. We include the bookkeeping because an unexplained cumulant truncation would not justify the main result. Use the same order as above, and set

$$V_i = \{j < i : j \sim i\}, \quad W_i = \{j < i : j \not\sim i\}.$$

In this proof all conditional quantities carrying subscript i are conditioned on F_{W_i} . Define

$$a_i = \sum_{j \in V_i} \mathbb{P}(E_j \mid F_{W_i}), \quad s_i = \sum_{j \in V_i} \mathbb{P}(E_i E_j \mid F_{W_i}).$$

Bonferroni's inequalities and independence of E_i from F_{W_i} give

$$\begin{aligned} \mathbb{P}(E_i \cap F_{V_i} \mid F_{W_i}) &= \pi_i - s_i + \rho_i, \\ \mathbb{P}(F_{V_i} \mid F_{W_i}) &= 1 - a_i + \eta_i, \end{aligned}$$

where

$$\begin{aligned} 0 \leq \rho_i &\leq e^{12\delta} \sum_{\{j,k\} \subset V_i} \pi_{ijk}, \\ 0 \leq \eta_i &\leq e^{8\delta} \sum_{\{j,k\} \subset V_i} \pi_{jk}, \quad \eta_i \leq a_i. \end{aligned}$$

These bounds follow from (20). Also $a_i \leq 2\delta_i \leq 1/4$, so the denominator is at least $3/4$. Expanding this quotient, with no infinite series, gives

$$z_i = \pi_i - s_i + \pi_i a_i + R_i, \quad |R_i| \leq C(\rho_i + a_i s_i + \pi_i a_i^2 + \pi_i \eta_i). \quad (26)$$

For completeness, subtract $\pi_i - s_i + \pi_i a_i$ from the quotient; the numerator is $\rho_i - s_i a_i - (\pi_i - s_i)\eta_i + \pi_i a_i^2 - \pi_i a_i \eta_i$. Use $\eta_i \leq a_i \leq 1/4$ and divide by at least $3/4$.

We now sum each error. The triples in ρ_i are connected stars, each counted at most three times, hence $\sum_i \rho_i \leq CT$. Further,

$$\sum_i a_i s_i \leq C\delta b_2, \quad \sum_i \pi_i a_i^2 \leq 4\delta b_1.$$

For $\sum_i \pi_i \eta_i$, split the pair $j, k \in V_i$ according to whether $j \sim k$. Nonadjacent pairs are independent, giving at most $C \sum_i \pi_i \delta_i^2 \leq C\delta b_1$. For adjacent pairs, first sum over $i \in B_j \cap B_k$: the sum of their π_i is at most δ . This gives at most $C\delta b_2$. Consequently

$$\sum_i |R_i| \leq C(T + \delta(b_1 + b_2)). \quad (27)$$

Next replace the conditional pair sum s_i by $v_i = \sum_{j \in V_i} \pi_{ij}$. The excess estimate (21), applied to $E_i E_j$, costs at most $C\delta \pi_{ij}$. The deficit estimate (22) costs at most $C \sum_{k \in W_i \cap B_j} \pi_{ijk}$: k cannot meet S_i because it belongs to W_i . These are connected three-vertex paths, and any unordered triple is counted at most six times. It follows that

$$\sum_i |s_i - v_i| \leq C(\delta b_2 + T). \quad (28)$$

Similarly replace a_i by $u_i = \sum_{j \in V_i} \pi_j$, remembering the factor π_i . The excess is bounded by $C\delta \sum_i \pi_i \sum_{j \in V_i} \pi_j \leq C\delta b_1$. The deficit is bounded by

$$C \sum_i \pi_i \sum_{j \in V_i} \sum_{k \in W_i \cap B_j} \pi_{jk}.$$

After summing over $i \in B_j$, this is at most $C\delta b_2$. Together with (26)–(28), this proves

$$\sum_i \left| z_i - \pi_i - \sum_{j \in V_i} (\pi_i \pi_j - \pi_{ij}) \right| \leq C(\delta(b_1 + b_2) + T). \quad (29)$$

It remains to pass from conditional probabilities to their logarithms. The first-order bounds in Lemma 3.1 imply

$$\sum_i |z_i - \pi_i| \leq C(b_1 + b_2).$$

Since $z_i + \pi_i \leq 3\pi_i \leq 3\delta$, $\sum_i |z_i^2 - \pi_i^2| \leq C\delta(b_1 + b_2)$. Also $\sum_i z_i^3 \leq 8\delta \sum_i \pi_i^2 \leq 8\delta b_1$. Taylor's formula on $[0, 1/4]$ now gives

$$\begin{aligned} \log \mathbb{P}(F_I) &= \sum_i \log(1 - z_i) \\ &= -\lambda + \sum_i \sum_{j \in V_i} (\pi_{ij} - \pi_i \pi_j) - \frac{1}{2} \sum_i \pi_i^2 + O(\delta(b_1 + b_2) + T). \end{aligned}$$

The middle two terms are $(\text{Var}(\sum_i \mathbf{1}_{E_i}) - \lambda)/2$: all covariances of nonadjacent events vanish. This proves the proposition. $\square$

4 Geometry of pairs and connected triples

Return to progression heads, with dependency graph defined by their supports. Throughout this section $r \geq 3$; absolute implied constants are sufficient. Put $\pi_i = \mathbb{P}(E_i) \leq p^r$.

4.1 Pair counts and compatible overlaps

Lemma 4.1 (Pair geometry; compare Zhao–Zhang, Lemma 3.2). *The number of ordered support-intersecting pairs is $O(n^3)$. The number with at least two shared support sites is $O(n^2 r^3)$. Among intersecting pairs with common differences in ratio two the number is $O(n^2)$. Every head has at most $O(nr)$ neighbors, and*

$$\delta \ll nrp^r, \quad (30)$$

$$b_1 + b_2 \ll n^3 p^{2r-1} + n^2 r^3 p^{5r/3-1} + n^2 p^{3r/2-1}. \quad (31)$$

For a compatible pair of distinct heads with different common differences d, e , put $m = \max(d/g, e/g)$, where $g = \gcd(d, e)$. Then

$$|S_i \cap S_j| \leq r/m + 1. \quad (32)$$

Distinct, intersecting heads with the same common difference are incompatible.

Proof. A fixed site belongs to at most $(r+1)\lfloor(n-1)/(r-1)\rfloor = O(n)$ supports: choose its index in $\{-1, 0, \dots, r-1\}$ and the common difference. As a support has at most $r+1$ sites, this gives $O(nr)$ neighbors. Since $|\mathcal{A}_{n,r}| = O(n^2/r)$, there are $O(n^3)$ ordered intersecting pairs.

If two supports share two sites, their common differences have ratio u/v for integers $1 \leq u, v \leq r$. Once the first head is fixed, there are at most r^2 choices of this ratio and at most $(r+1)^2$ choices of indices specifying a common point. Hence there are $O(r^4)$ such second heads and $O(n^2 r^3)$ such pairs. For $e = 2d$, a common point forces $b = a + kd$ with k in an interval of length $O(r)$; there are only $O(r)$ second starts for each first head. Interchanging d, e proves the $O(n^2)$ estimate.

Two common support points differ by a multiple of $\text{lcm}(d, e)$. Their indices in the two supports therefore differ by multiples of e/g and d/g , respectively. The available index spans are at most r , which proves (32). When $d = e$ and $b > a$, an intersection gives $b = a + kd$ for $1 \leq k \leq r$. The later head requires $\xi_{b-a} = 0$, while that site belongs to the earlier positive core. The intersection event is empty.

A one-site-overlap pair has at least $2r - 1$ distinct required positive sites, and its joint probability is at most p^{2r-1} . For different differences in ratio two it is at most $p^{3r/2-1}$. For all other compatible multiply-overlapping pairs, $m \geq 3$ and the joint probability is at most $p^{5r/3-1}$. Multiplying these bounds by the counts proves the b_2 estimate. The b_1 estimate $O(n^3 p^{2r})$ is absorbed by the first term, and (30) follows from the degree bound. $\square$

4.2 A triple estimate adequate for a second-order law

Lemma 4.2 (Connected triples). *For the T in (18),*

$$T \ll n^4 r^4 p^{3r-3} + n^3 r^5 p^{5r/2-3} + n^2 r^8 p^{5r/3-3}. \quad (33)$$

Proof. A triple contributes only if the three head requirements are compatible. Classify its pair edges as multiple when at least two support sites are shared. There are three cases.

No multiple edges. Choose a root head and then attach two heads along a spanning tree. There are $O(n^2)$ choices for the root and $O(nr)$ choices for each attachment, giving at most $O(n^4 r^4)$ triples

(the powers used here are deliberately loose). Each pair of positive cores intersects in at most one site, so the union of the three cores has at least $3r - 3$ sites.

Exactly one multiple edge. Choose that pair in $O(n^2 r^3)$ ways. The third head meets at least one member, giving $O(nr)$ choices, hence at most $O(n^3 r^5)$ triples. The multiple pair, if compatible, has intersection at most $r/2 + 1$; each other intersection has size at most one. The positive union has size at least $5r/2 - 3$.

At least two multiple edges. Two multiple edges contain a spanning tree. Starting from a root, each attachment has $O(r^4)$ choices by the two-site argument. There are at most $O(n^2 r^8)$ triples. If all three common differences are distinct, every pair overlap is at most $r/2 + 1$, and at least one is at most $r/3 + 1$. Indeed three distinct positive integers cannot have every pair in ratio two. Inclusion–exclusion therefore bounds the positive union below by $5r/3 - 3$. If two common differences coincide, compatibility forces their supports to be disjoint. The other two overlaps are at most $r/2 + 1$, giving the stronger lower bound $2r - 2$. If all coincide, connectedness and compatibility cannot both hold.

In each case ignore required zeros, multiply the count by p to the appropriate lower bound on the positive union, and sum. This gives the stated estimate even when a displayed fractional exponent is negative for small r , since such an upper bound remains valid. $\square$

5 The entropy profile of progression incidence

For $x \in [n]$, let $d_r(x)$ be the number of r -term positive cores containing x . The main coefficient comes from these degrees, not from the rare high-multiplicity intersections.

Lemma 5.1 (Uniform degree profile). *Let $h(u) = -u \log u - (1-u) \log(1-u)$, with $h(0) = h(1) = 0$. Uniformly in $x \in [n]$, for $3 \leq r \leq n$,*

$$d_r(x) = nh \left(\frac{x-1}{n-1} \right) + O \left(\frac{n}{r} + r \right). \quad (34)$$

Consequently, uniformly for $3 \leq r \leq n$,

$$\sum_{x=1}^n d_r(x)^2 = n^3 H + O \left(\frac{n^3}{r} + n^2 r \right). \quad (35)$$

Proof. Put $m = r - 1$ and $u = (x - 1)/(n - 1)$. If x is at position j in its progression, $0 \leq j \leq m$, the permissible differences are the positive integers at most

$$\min \left(\frac{x-1}{j}, \frac{n-x}{m-j} \right),$$

where an endpoint's missing constraint is omitted. Thus

$$d_r(x) = \frac{n-1}{m} \sum_{j=0}^m f_u(j/m) + O(r), \quad f_u(t) = \min \left(\frac{u}{t}, \frac{1-u}{1-t} \right),$$

with $f_u(0) = 1 - u$ and $f_u(1) = u$. The function increases up to its peak at $t = u$ and then decreases, has height at most one, and has total variation at most two, including $u = 0, 1$. Therefore its Riemann sum satisfies

$$\frac{1}{m} \sum_{j=0}^m f_u(j/m) = \int_0^1 f_u(t) dt + O(1/m)$$

uniformly in u . Splitting the integral at u gives

$$\int_0^u \frac{1-u}{1-t} dt + \int_u^1 \frac{u}{t} dt = h(u).$$

This proves (34).

The function h^2 is bounded and has bounded variation, so its Riemann sum on the n grid points is $nH + O(1)$. Squaring the degree formula and summing contributes an error at most

$$O\left(n^2 \left(\frac{n}{r} + r\right) + n \left(\frac{n}{r} + r\right)^2 + n^2\right).$$

For $3 \leq r \leq n$ this is $O(n^3/r + n^2r)$, proving (35). In particular, $n^{-3} \sum_x d_r(x)^2 \rightarrow H$ whenever $r \rightarrow \infty$ and $r = o(n)$. $\square$

Lemma 5.2 (Evaluation of the entropy integral). *The value of H is $5/6 - \pi^2/18$.*

Proof. Expansion of the square and symmetry give

$$H = 2 \int_0^1 u^2 \log^2 u \, du + 2 \int_0^1 u(1-u) \log u \log(1-u) \, du.$$

The first integral is $2/27$. Expanding $-\log(1-u)$ in its nonnegative power series and integrating by monotone convergence evaluates the second integral as

$$J = \sum_{k \geq 1} \frac{1}{k} \left(\frac{1}{(k+2)^2} - \frac{1}{(k+3)^2} \right).$$

For an integer $a \geq 1$, partial fractions give

$$\sum_{k \geq 1} \frac{1}{k(k+a)^2} = \frac{H_a}{a^2} - \frac{\zeta(2) - H_a^{(2)}}{a},$$

where $H_a = \sum_{j=1}^a 1/j$ and $H_a^{(2)} = \sum_{j=1}^a 1/j^2$. Taking the case $a = 2$ minus the case $a = 3$ yields $J = 37/108 - \pi^2/36$. Hence $H = 4/27 + 2J = 5/6 - \pi^2/18$. $\square$

6 Computing the leading covariance term

For a head i and a coordinate x , define its signed influence by

$$D_{i,x} = \mathbb{E}(\mathbf{1}_{E_i} \mid \xi_x = 1) - \mathbb{E}(\mathbf{1}_{E_i} \mid \xi_x = 0).$$

It is zero outside S_i . If x belongs to the positive core it equals p^{r-1} for a boundary head and qp^{r-1} for an interior head. At the predecessor of an interior head it equals $-p^r$.

Lemma 6.1 (One-site covariance identity). *If $S_i \cap S_j = \{x\}$, then*

$$\text{Cov}(\mathbf{1}_{E_i}, \mathbf{1}_{E_j}) = pq D_{i,x} D_{j,x}. \quad (36)$$

Let $B_r(x)$ count boundary heads whose positive core contains x , and put $P_r(x) = \lfloor (n-x)/r \rfloor$. Then, exactly,

$$S(x) := \sum_i D_{i,x} = qp^{r-1} d_r(x) + p^r B_r(x) - p^r P_r(x), \quad B_r(x), P_r(x) \leq \frac{n}{r-1}. \quad (37)$$

Proof. Given ξ_x , the remaining coordinates of the two events are disjoint and hence independent. The covariance of two affine functions of a Bernoulli- p variable is pq times the product of their slopes, proving (36). The formula for $S(x)$ follows from the three influence values above. A head with predecessor x has start $x + d$ and endpoint $x + rd \leq n$, so there are exactly $P_r(x)$ such heads. For each fixed common difference d , at most one boundary start $a \in \{1, \dots, d\}$ can have x in its core: it is the unique representative of x modulo d in that interval. This bounds $B_r(x)$ by D_r . $\square$

Proposition 6.2 (Second factorial cumulant). *Fix $p \in (0, 1)$, and suppose $r \rightarrow \infty$ with $r = O(\log n)$. Then*

$$\frac{\text{Var } Y - \lambda}{2} = \frac{q^3 H}{2p} n^3 p^{2r} + O_p(V_{n,r}), \quad (38)$$

$$V_{n,r} = \frac{n^3 p^{2r}}{r} + n^2 r p^{2r} + n^2 r^4 p^{2r-2} + n^2 p^{3r/2-1} + n^2 r^3 p^{5r/3-1}. \quad (39)$$

Proof. By (37), $S(x) = qp^{r-1}d_r(x) + O(p^r n/r)$. Using Lemma 5.1 and $d_r(x) = O(n)$ yields

$$\frac{pq}{2} \sum_x S(x)^2 = \frac{q^3 H}{2p} n^3 p^{2r} + O_p \left(\frac{n^3 p^{2r}}{r} + n^2 r p^{2r} \right). \quad (40)$$

Expanding the square on the left gives a diagonal sum over individual heads plus sums over unordered pairs. By (36), the one-site-overlap pairs have exactly their true covariance contribution. The diagonal projection has size $O_p(n^2 p^{2r-2})$: there are $O(n^2/r)$ heads and $O(r)$ nonzero influences per head. The projection contributions of multiply-overlapping pairs are at most

$$O_p((n^2 r^3) r p^{2r-2}) = O_p(n^2 r^4 p^{2r-2}).$$

Their actual covariance sum in absolute value is bounded by their joint probability sum plus the product-probability sum. Lemma 4.1 therefore bounds it by

$$O \left(n^2 p^{3r/2-1} + n^2 r^3 p^{5r/3-1} + n^2 r^3 p^{2r} \right).$$

Finally, the second factorial cumulant subtracts $\frac{1}{2} \sum_i \pi_i^2 = O(n^2 p^{2r}/r)$ from the unordered covariance sum. All diagonal and product terms are absorbed by the third term of (39). Combining these observations proves the proposition. $\square$

Remark 6.3 (Why the sign is positive). The dominant term is a sum of squares of aggregate signed influences. Although predecessors introduce negative influences and same-difference heads can exclude one another, their aggregate contribution is smaller by a factor of at least $1/r$ in (37). The leading term is therefore the positive entropy integral. Positivity is a conclusion of this calculation, not an application of positive association to the nonmonotone head events.

7 Assembly of the critical and uniform expansions

7.1 A logarithmic band with a weighted remainder

Lemma 7.1 (Uniform expansion on a logarithmic band). *Fix positive constants c_1, c_2, C_0 . Suppose*

$$c_1 \log n \leq r \leq c_2 \log n, \quad 0 < \lambda = \lambda_{n,r} \leq C_0 \log n.$$

Then, for all sufficiently large n ,

$$\log \mathbb{P}(U_n < r) = -\lambda + c_p \lambda^2 \frac{r^2}{n} + R_{n,r}, \quad |R_{n,r}| \ll_{p,c_1,c_2,C_0} (\lambda^{3/2} + \lambda^3) \frac{r^{3/2}}{n}. \quad (41)$$

Proof. On this band (16) gives $p^r \asymp_p \lambda r/n^2$. In particular $\delta \ll_p \lambda r^2/n = O((\log n)^3/n)$, so the local avoidance lemmas apply. The pair and triple estimates become

$$b_1 + b_2 \ll_p \lambda^2 \frac{r^2}{n} + \lambda^{5/3} \frac{r^{14/3}}{n^{4/3}} + \lambda^{3/2} \frac{r^{3/2}}{n}, \quad (42)$$

$$T \ll_p \lambda^3 \frac{r^7}{n^2} + \lambda^{5/2} \frac{r^{15/2}}{n^2} + \lambda^{5/3} \frac{r^{29/3}}{n^{4/3}}. \quad (43)$$

Also (16) converts the main covariance term into

$$\frac{q^3 H}{2p} n^3 p^{2r} = c_p \lambda^2 \frac{r^2}{n} \left(1 + O_p \left(\frac{1}{r} + \frac{r}{n} \right) \right). \quad (44)$$

Substituting into $V_{n,r}$ in (39) produces terms bounded by

$$O_p \left(\lambda^2 \frac{r}{n} + \lambda^2 \frac{r^6}{n^2} + \lambda^{3/2} \frac{r^{3/2}}{n} + \lambda^{5/3} \frac{r^{14/3}}{n^{4/3}} \right).$$

The error in (44) is absorbed in these terms. Combining Proposition 3.3 with Proposition 6.2, including the product $\delta(b_1 + b_2)$ from (42), bounds the total remainder by

$$C_p (\lambda^{3/2} + \lambda^3) \left(\frac{r^{3/2}}{n} + \frac{r^{10}}{n^{4/3}} + \frac{r^8}{n^2} \right). \quad (45)$$

To see that the powers of λ suffice, every exponent arising here is between $3/2$ and 3 ; $\lambda^s \leq \lambda^{3/2} + \lambda^3$ in that range. Since $r \asymp \log n$, the last two terms in parentheses are $o(r^{3/2}/n)$, uniformly on the band. This proves (41). $\square$

Proof of Theorem 1.2. First, $\lambda \asymp 1$ forces $r \asymp \log n$. Indeed the elementary upper bound $\lambda \leq n^2 p^r$ gives $r = O(\log n)$, whereas $\lambda \geq qp^r M_r(n)$ and (1) give a matching lower bound once $r = O(\log n)$. Fixed r cannot keep λ bounded as $n \rightarrow \infty$. Taking logarithms in (16) then yields (6). Lemma 7.1 proves (7). Exponentiation proves (8), because $r^4/n^2 = o(r^{3/2}/n)$. Finally (15) is $O_{p,a,b}(r/n)$ in this window, smaller than the stated remainder. The same relative mean estimate shows that a critical condition on $\bar{\lambda}$ is equivalent to a slightly enlarged critical condition on λ . $\square$

7.2 Why the tails do not lose logarithms

Proof of Theorem 1.1 for the exact mean. Put $L = \log n$. Let r_- be the smallest $r \geq 3$ with $\lambda_{n,r} \leq (16/p)L$, and let $r_+ = \lceil 3L/\alpha \rceil$. For large n , both are of order L , $r_- < r_+$, and

$$8L \leq \lambda_{n,r_-} \leq (16/p)L. \quad (46)$$

To justify the lower inequality, uniformly for $r \asymp L$ the explicit mean formula gives $\lambda_{n,r}/\lambda_{n,r-1} = p(1 + O_p(1/r + r/n)) \geq p/2$. The defining previous mean is larger than $(16/p)L$.

For $r_- \leq r \leq r_+$, the hypotheses of Lemma 7.1 hold. Its correction $K = c_p \lambda^2 r^2/n$ and remainder R are both uniformly $o(1)$, since $\lambda = O_p(L)$. Hence

$$\begin{aligned} \left| \mathbb{P}(U_n < r) - e^{-\lambda}(1 + K) \right| &\ll_p e^{-\lambda} \left((\lambda^{3/2} + \lambda^3) \frac{r^{3/2}}{n} + \lambda^4 \frac{r^4}{n^2} \right) \\ &\ll_p \frac{L^{3/2}}{n}. \end{aligned}$$

The last inequality uses boundedness of $x^s e^{-x}$ on $[0, \infty)$. This exponential damping is precisely what is lost in a uniform proof that takes the worst overlap bound before controlling the avoidance tail.

For $r < r_-$, (42) implies $b_2 = o(\lambda)$ at r_- . Lemma 3.2 and (46) give $\mathbb{P}(U_n < r_-) \leq e^{-\lambda_{n,r_-}/2} \leq n^{-4}$. Both $\mathbb{P}(U_n < r)$ and the uncorrected exponential are then negligible by monotonicity. The correction is negligible as well: here $r \leq r_- = O_p(L)$ and $\lambda \geq 8L$, so $e^{-\lambda} \lambda^2 r^2 / n = O_p(L^4 n^{-9})$.

For $r > r_+$, Markov's inequality and Lemma 2.1 give $\mathbb{P}(U_n \geq r) \leq \lambda_{n,r} \leq \lambda_{n,r_+} = O_p(1/(nL))$. Also $1 - e^{-\lambda} \leq \lambda$. Finally the global bound $\lambda \ll n^2 p^r / r$ gives

$$e^{-\lambda} \lambda^2 r^2 / n \ll n^3 p^{2r} \leq n^{-3}.$$

These estimates settle the upper tail and complete the exact-mean proof. $\square$

Transfer to the smooth mean. On the middle band, (15) gives $|\bar{\lambda} - \lambda| \ll_p \lambda r / n$ and $\bar{\lambda} / \lambda = 1 + O_p(r/n)$. Apply the mean value theorem to $g_r(x) = e^{-x}(1 + c_p x^2 r^2 / n)$. Its derivative is

$$g'_r(x) = e^{-x} \left[-1 + c_p \frac{r^2}{n} (2x - x^2) \right].$$

On the segment between the two means, the exponential is at most $e^{-\lambda/2}$ for sufficiently large n . The resulting difference is $O_p(L/n)$ uniformly, by exponential damping. In the lower tail the relative mean estimate is still valid for $2 \leq r < r_-$, using (1) and $r = O_p(L)$; thus $\bar{\lambda} \geq \lambda/2 \geq 4L$. Both smooth terms are negligible. In the upper tail $\bar{\lambda} \ll_p n^2 p^r / r$, so the same estimates as above apply. This proves (5) for $\bar{\lambda}$. $\square$

8 The optimal error has an explicit periodic amplitude

Proof of Theorem 1.3. Theorem 1.1, and the elementary inequality $||x| - |y|| \leq |x - y|$, imply

$$\mathcal{E}_n(\lambda) = \frac{c_p}{n} \max_{2 \leq r \leq n} \left(e^{-\lambda_{n,r}} \lambda_{n,r}^2 r^2 \right) + O_p(L^{3/2}/n). \quad (47)$$

Let $f(x) = x^2 e^{-x}$. For $r = \lfloor t_n \rfloor + j$ with j in any fixed finite set, (16) gives, uniformly,

$$\lambda_{n,r} = e^{\alpha(\{t_n\} - j)} \left(1 + O_p\left(\frac{\log L}{L}\right) \right), \quad r^2 = \frac{4L^2}{\alpha^2} + O_p(L \log L). \quad (48)$$

The constant in t_n is determined by $qn^2 e^{-\alpha t_n} / (2(2L/\alpha)) = 1$.

We justify restricting the maximum to a fixed finite set of such j . A lattice point with its limiting mean in $[p, 1]$ provides $f(\lambda) \geq m_p > 0$ for all sufficiently large n . The two tails $r < r_-$ and $r > r_+$ in the preceding proof contribute $o(L^2)$ to the maximum in (47). On the middle band r^2/L^2 is bounded above and below by positive constants. Since $f(x) \rightarrow 0$ at zero and infinity, one can choose $0 < a_p < b_p < \infty$ so that a maximizer must have $a_p \leq \lambda_{n,r} \leq b_p$. Formula (6), with its more precise constant supplied by (16), then gives $r - t_n = O_p(1)$. The analogous maximum defining Φ_p also lies in a fixed finite set of j , uniformly for $\theta \in [0, 1]$.

The derivative of f is bounded on the relevant compact interval. Thus (48) converts (47) to

$$\mathcal{E}_n(\lambda) = \frac{4c_p L^2}{\alpha^2 n} \Phi_p(\{t_n\}) + O_p\left(\frac{L \log L}{n} + \frac{L^{3/2}}{n}\right).$$

Since $L \log L = O(L^{3/2})$, this is (10). The smooth parameter obeys the same lattice-mean expansion, so the same argument applies directly to $\mathcal{E}_n(\bar{\lambda})$.

It remains to evaluate the extrema of Φ_p . The unrestricted maximum of f is $f(2) = 4/e^2$, so $\max_{\theta} \Phi_p(\theta) = 4/e^2$. For a multiplicative lattice of ratio p , only the two adjacent points x and px straddling 2 can maximize f . Thus $2 \leq x \leq 2/p$, and its maximum is $\max(f(x), f(px))$. The first term decreases and the second increases on that interval. Their equality occurs at

$$x_* = -\frac{2 \log p}{1-p} = \frac{2\alpha}{q} \in (2, 2/p).$$

Therefore

$$\min_{\theta} \Phi_p(\theta) = \frac{4\alpha^2}{q^2} e^{-2\alpha/q} > 0, \quad \max_{\theta} \Phi_p(\theta) = \frac{4}{e^2}. \quad (49)$$

The maximum over a uniformly finite set also proves continuity of Φ_p , and relabeling j proves periodicity.

Finally $t_n \rightarrow \infty$, is eventually increasing, and $t_{n+1} - t_n \rightarrow 0$. For every $\theta \in [0, 1]$, take the first integer n crossing each successive level $k + \theta$. The overshoot tends to zero, so the fractional parts $\{t_n\}$ have θ as an accumulation point (with the endpoints understood periodically). Equations (10) and (49) now prove (11) and (12). They are distinct because $x_* > 2$ and f is strictly decreasing on $(2, \infty)$. $\square$

Remark 8.1 (Fair-coin specialization). At $p = q = 1/2$, the normalized error has

$$\liminf \frac{n\mathcal{E}_n}{(\log n)^2} = 4c_{1/2} \approx 2.280175821738063, \quad \limsup \frac{n\mathcal{E}_n}{(\log n)^2} = \frac{16c_{1/2}}{(\log 2)^2 e^2}.$$

The oscillation is not a numerical fitting phenomenon. It comes from the integer spacing of progression lengths and the geometric spacing of their Poisson means.

9 A separate sparse-density consequence

The preceding main theorems fix p and let r grow. The head viewpoint also gives an elementary consequence in the opposite regime, useful for cross-checking what declumping does and does not remove.

Proposition 9.1 (Fixed length, sparse density). *Fix $k \geq 3$ and $C_0 < \infty$. Uniformly for $0 \leq c \leq C_0$ and $p = cn^{-2/k}$ (for all sufficiently large n),*

$$\mathbb{P}(U_n < k) = \exp(-M_k(n)p^k + M_{k+1}(n)p^{k+1}) + O_{k,C_0}(n^{-1+2/k}). \quad (50)$$

For each fixed $c > 0$ and $k \geq 5$, this implies

$$\log \mathbb{P}(U_n < k) = -\frac{c^k}{2(k-1)} + \frac{c^{k+1}}{2k} n^{-2/k} + O_{k,c}(n^{-1+2/k}). \quad (51)$$

Proof. With k fixed, there are $O_k(n^3)$ intersecting support pairs and $O_k(n^2)$ multiply-overlapping pairs. A pair of cores with different common differences shares at most $\lceil k/2 \rceil$ positive sites: the common index increments have maximum at least two. Same-difference intersecting heads are incompatible. Thus

$$b_1 \ll_k n^3 p^{2k}, \quad b_2 \ll_k n^3 p^{2k-1} + n^2 p^{2k-\lceil k/2 \rceil}.$$

At $p = cn^{-2/k}$ these are $O_{k,C_0}(n^{-1+2/k})$; for the second term the exponents are -1 for even k and $-1 + 1/k$ for odd k , both smaller. Also $\delta \ll_k np^k = O_{k,C_0}(n^{-1})$. Lemma 3.2 gives the claimed approximation, without requiring a lower bound on c ; the case $c = 0$ is exact. For fixed $c > 0$ the avoidance probability stays bounded away from zero. Taking logarithms and using $M_j(n) = n^2/(2(j-1)) + O_j(n)$ proves (51). For $k \geq 5$ its displayed correction exceeds the remainder, since $2/k < 1 - 2/k$. $\square$

Remark 9.2 (Interpretation, not a priority claim). The exact mean incorporates the first sliding-run correction $M_{k+1}p^{k+1}$. This is an illustration of declumping, not a claim that fixed-pattern avoidance expansions originate here; the more general fixed-length cumulant literature includes [2]. The genuinely separate issue in the main theorems is uniformity as $r \asymp \log n$, together with evaluation and sharpness of the resulting leading error.

10 Verification, limitations, and integration

10.1 What the executable artifact actually checks

The supplied `code/verify.py` uses only Python’s standard library. It enumerates Bernoulli configurations and uses integers and rational numbers for the finite event identities. The recorded full run passed:

Check family	Number checked
Configuration/threshold instances	114,176
Exhaustive (n, r) cases	62
Pair parameter cases, including three rational densities	36
One-site covariance identities	21,183
Multiply-overlapping pair classifications	9,546
Three-head geometric configurations	810,589
Compatible connected triples among these	194,077
Numerical local-avoidance sandwiches	8

The configuration checks verify the exact zero-event reduction, the mean formula, and selected exact second moments. Pair checks also verify the signed-influence sum and the projection decomposition. Triple checks verify the three positive-union lower bounds used in Lemma 4.2. The eight logarithmic sandwich checks use floating-point logarithms of exact rational probabilities and are labeled numerical, rather than exact inequalities.

The degree-profile calculation checks the exact identity $\sum_x d_r(x) = rM_r(n)$ and records $n^{-3} \sum_x d_r(x)^2$ for n up to 100,000. Its agreement with the entropy limit is a diagnostic, not evidence sufficient to establish an asymptotic theorem. The exact configuration enumeration is intentionally small. It neither estimates a tiny large- n correction by Monte Carlo nor silently treats finite checks as a proof for all n .

10.2 Claims that are not being made

No theorem in this paper is currently certified by a proof assistant. The manuscript’s written proofs, especially Proposition 3.3 and its weighted use in Lemma 7.1, require independent mathematical review. The novelty comparison is specific to the inspected sources and is not an exhaustive priority determination. The remainder $O((\log n)^{3/2}/n)$ after correction is an upper bound; we do not prove

it is optimal. No uniformity as p approaches zero or one follows from the fixed- p theorems. No claim is made for cyclic progressions modulo n , for a fixed-cardinality random set, or for an improvement of deterministic Ramsey numbers.

10.3 Repository placement and proof dependencies

A suggested destination, not an assertion that the folder already exists, is

Combinatorics/Ramsey/Research/RandomProgressions/SharpPoisson/.

The package contains this manuscript, its source, an external bibliography, reproducible code, recorded data, a claims ledger, a literature/novelty note, and a formalization roadmap. It does not modify either repository. The proof dependencies are

$$\begin{aligned}
 &\text{conditional control} \implies \text{local avoidance expansion}, \\
 &\left. \begin{array}{l} \text{local avoidance expansion} \\ \text{pair and triple geometry} \\ \text{entropy covariance calculation} \end{array} \right\} \implies \text{weighted band expansion}, \\
 &\text{weighted band expansion} \implies \text{uniform correction} \implies \text{lattice error law}.
 \end{aligned}$$

11 Further research questions

The following questions are not used in the proofs above.

1. Evaluate the ratio-two correction. Pairs with common differences in ratio two account for the visible $O(r^{3/2}/n)$ term in the critical window. Determine their exact contribution, including dependence on the parity of r and on the lattice phase of n . Does subtracting it reduce the uniform remainder to $O((\log n)/n)$? A solution would need actual covariance counts, not just the upper bound in Lemma 4.1.

2. A third-order avoidance expansion with an effective constant. Extend Proposition 3.3 by one order for nonmonotone cylinder events. Separate the connected four-event terms from products of pair terms, and supply explicit numerical constants valid under a stated local-mass hypothesis. Such a result would support a systematic expansion rather than an isolated correction.

3. Uniform regimes for varying $p = p_n$. Determine the largest ranges in which the entropy correction remains dominant when $p_n \rightarrow 0$ or $p_n \rightarrow 1$. The present error terms contain powers of p^{-1} and q^{-1} after mean normalization, so simply replacing p by p_n in the main theorems is not justified. Locate the transition between growing-pattern behavior and fixed-pattern sparse avoidance.

4. Conditioning on the cardinality of the random set. For a uniformly random $\lfloor pn \rfloor$ -element subset, the total number of selected coordinates no longer fluctuates. The first projection suggests centering the profile h by its integral $1/2$, replacing H by $H - 1/4$ in one part of the calculation. This is only a heuristic: a proof must also account for the global negative dependence introduced by sampling without replacement.

5. Cyclic progressions and arithmetic obstructions. Determine the corrected avoidance law for progressions modulo n . The interval degree profile is then replaced by a different incidence geometry, and short cycles or small divisors of n can affect overlap counts. A result uniform in composite n will require explicit handling of these arithmetic obstructions.

6. Multicolor versions. For independent random colorings with prescribed color probabilities, study the longest monochromatic progression and the joint head-count vector by color. Same-site influences become vectors, and the relevant coefficient should be a quadratic form rather than a single scalar. Identify when cross-color exclusions contribute at leading order.

7. Waiting times and moments. Let $\tau_r = \min\{n : U_n \geq r\}$. Transfer the corrected distribution to precise asymptotics for its quantiles, expectation, and variance. Moment asymptotics require tail integration estimates beyond a local inversion of Theorem 1.2; the lattice effects must not be averaged away without justification.

8. Kernel-checked formalization. Formalize the finite head bijection, same-difference incompatibility, two-site rational-ratio lemma, and the one-coordinate covariance identity first. Then formalize conditional local-lemma bounds and the finite Bonferroni quotient expansion. The analytic degree-profile limit and the lattice-maximization step can be separate modules. The code supplied here is a regression suite, not a substitute for these proofs.

Conclusion

The precise error of the longest-progression Poisson law is governed by a geometric quantity that is invisible in an unweighted dependency-graph bound: the squared entropy profile of progression incidence. After separating the one-site projection from rare multiple overlaps, the second factorial cumulant has an explicit positive coefficient. A local avoidance expansion with a controlled connected-triple remainder then turns this coefficient into a uniform probability correction. The resulting order $(\log n)^2/n$ is sharp, and its leading constant oscillates with a fully specified lattice phase. The next unresolved scale in this analysis is the ratio-two contribution of order at most $(\log n)^{3/2}/n$.

References

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